



The Federal University of Technology, Akure.
School of Engineering and Engineering Technology
Department of Information & Communication Technology
Second Semester Examinations 2023/2024 Session

ICT 506 Information Systems Analysis and Design **Date: 23rd September, 2024.**
Degree: B. Eng. Information and Communication Technology

Instructions:

ANSWER Any Three Questions in all.

Time Allowed: 2 HOURS.

Question 1

- a. Describe the major phases in the systems development life cycle (SDLC). **6 marks** *Chapter 1 Q5*
- b. List six general skills all project team members should have. **3 marks** *Chapter 1 Q1*
- c. Consider the Amazon.com website. The management of the company decided to extend its Web-based system to include products other than books (e.g., wine, specialty gifts). *Chapter 1 EF*
- (i) How would you have assessed the feasibility of this venture when the idea first came up? **3 marks**
- (ii) How "risky" would you have considered the project that implemented this idea? **2 marks**
- d. Compare and contrast the role of a systems analyst, business analyst, and infrastructure analyst. **6 marks** *Chapter 1 Q3*

Question 2

- a. Suppose that you are an analyst developing a new executive information system (EIS) intended to provide key strategic information from existing corporate databases to senior executives to help in their decision making. Explain with reasons the methodology you would use. **5 marks** *Chapter 2 E3*
- b. Explain the meaning of analysis and state the purpose of the analysis phase of the SDLC. **5 marks** *Chapter 3 Q1*
- c. State the three basic steps of the analysis process and explain why each step may not perform in every project **5 marks** *Chapter 3 Q2*
- d. (i) Compare and contrast outcome analysis, technology analysis, and activity elimination. **3 marks**
- (ii) What general contribution do these strategies play in determining requirements? **2 marks**

Question 3

- a. Explain the term *entity* in a data model and state three steps in building entity relationship diagrams. **5 marks** *Chapter 6*
- b. Summarize the distinctions between the analysis phase and the design phase of the SDLC. **5 marks** *Chapter 7 Q1*
- c. Describe the primary activities of the design phase of the SDLC. **5 marks** *Chapter 7 Q2*
- d. Assume that you are developing a new system for a local real estate agency. The agency wants to keep a database of its own property listings and also wants to have access to the citywide multiple listings service used by all real estate agents. Which design strategy would you recommend for the construction of this system and explain why? **5 marks** *Chapter 8 Q1*

*packaged
market software*

Question 4

- a. Explain briefly, the difference between a methodology and a work plan and state how they are related.
Chapter 2 Q15 **5 marks**
- b. Describe how projects are selected in organizations. *Chapter 2 Q1* **5 marks**
- c. Some companies hire consulting firms to develop the initial project plans and manage the project but use their own analysts and programmers to develop the system. Explain why some companies do this?
Chapter 2 Q16 **5 marks**
- d. Enumerate the reasons why many projects end up having unreasonable deadlines and explain how should a project manager react to unreasonable demands?
Chapter 2 Q15 **5 marks**

Question 5 ✓

interview, questionnaires, new, known joint application development.

- a. Suppose that you are the analyst charged with developing a new system to help senior managers make better strategic decisions. What requirements gathering techniques will you use and describe in detail how you would apply the techniques. **6 marks**
- b. Explain process model and data flow diagram. State the relationship between them. **5 marks**
- c. Explain the following as applicable to Process Model: (i) Process (ii) Data flow (iii) data store (iv) External entity *Q3 Q4 Q5* **4 marks**
- d. Describe a "risky" project in terms of technical feasibility and explain a project that would *not* be considered risky. *Chapter 1 - Q23* **5 marks**

*Process model
graphical representation
of the business model.*

data flow diagram - specific type of process model that focuses on data flows and transformation.

*c) Process: Activities performed in the business process.
Data flow: use of data or collection of data element that flows between processes, data stores, &*

*2 external entities
Data stores: collection of stored data*

External entity: person or organization outside the system model



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COURSE CODE: ICT506 **COURSE TITLE:** Information Systems Analysis and Design

UNITS: 2.0

TIME ALLOWED: 2 Hours

Instruction: Read Carefully and Answer any three (3) questions

Question 1

- a. A project at "Tune Source" involves developing a brand-new, complex music recommendation system with initial requirements that are not fully defined. As the project manager:
- (i) discuss in details how you would justify your selection of a specific project methodology such as Agile or Waterfall. (5 ½ marks)
 - (ii) Explain how this choice would influence the process of Estimating the Project Time Frame and Developing the Work Plan. (5 marks)
- b. Discuss in details how the use of rapid application development tools affects the traditional System Development Life Cycle phases of logical and physical design. (5 ½ marks)
- c. Explain the following as applicable to Process Model: (i) Process (ii) Data flow (iii) data store (iv) External entity (1 mark each)

Question 2

- a. (i) Discuss the major cause of severe uncontrolled expansion of a project's scope within a systems development project and elaborate on the specific, quantifiable impacts this would typically have on the project's budget and timeline. (5 marks)
- (ii) Identify and justify three specific techniques for Managing Scope and two techniques for Refining Estimates that you would utilize to effectively mitigate the impacts and bring the project back under control. (5 marks)
- b. Suppose that you are the analyst charged with developing a new system to help senior managers make better strategic decisions. What requirements gathering techniques will you use and describe in detail how you would apply the techniques. (5 marks)
- c. Describe a scenario where an organization would prefer to use a traditional, structured System Development Life Cycle approach over a rapid application development methodology. (5 marks)

Question 3

- a. You are a project manager responsible for Staffing a Project and a key team member unexpectedly leaves.
- (i) Enumerate how you would Coordinating the Project Activities to mitigate the risks associated with this staffing change. (5 marks)
 - (ii) Discuss the role of a detailed Work Plan and a proactive Risk Management strategy in this type of scenario (5 marks)
- b. As an expert in systems development life cycle, explain the reason behind some companies hiring consulting firms to develop the initial project plans and manage the project but prefer to use their own analysts and programmers to develop the system. (5 marks)



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- c. Explain the relationship between the techniques of Data Flow Diagramming and Entity-Relationship Diagramming within the Systems Development Life Cycle. (5 marks)

Question 4

- a. Compare and contrast Interviews and Joint Application Development (JAD) requirements Elicitation Techniques in systems analysis development life cycle. (10 marks)
- b. Discuss a specific project scenario that would determine the choice of one technique over the other and state the factors that would influence your decision. (5 marks)
- c. Explain with reasons the methodologies you would adopt as an analyst contracted to develop a new executive information system (EIS) intended to provide key strategic information from existing corporate databases to senior executives in an organization to help them in their decision making. (5 marks)

Question 5

- a. (i) Explain how a systems analyst could use a combination of Root Cause Analysis and Activity-Based Costing to improve an inefficient business process. (4 marks)
- (ii) Provide a specific example of a problem and walk through how these two analysis strategies would be applied to identify the underlying issues and justify the proposed solution. (4 marks)
- b. Explain the main purpose of the Maintenance phase in Systems Development Life Cycle and list the two types of maintenance. (4 marks)
- c. A key challenge in Requirements Determination is translating business needs into a clear and verifiable Requirements Definition Statement.
- (i) As a systems analyst, provide your approach to ensuring a requirement is properly defined and not ambiguous. (4 marks)
- (ii) Discuss the importance of using Document Analysis and Observation techniques to validate information to prevent the inclusion of unnecessary or conflicting requirements. (4 marks)